The natural side of the comparison

Influenza A hemagglutinin subtypes, sequence databases, redundancy weighting and alignment

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Abstract

The sampler produces a per-site entropy over the sequences it accepts. To say whether that entropy means anything we compare it with the per-site entropy of real hemagglutinin (HA) sequences. Until now the natural side was 19 380 human H3N2 sequences; this note explains what we are replacing that with and why, and defines every piece of jargon in the plan: subtype groups, influenza B, the NCBI, BV-BRC and GISAID databases, CD-HIT and MMseqs2 clustering, DCA-style sequence weights, and MAFFT alignment. Section 7 is the long version of the one real judgement call in the pipeline.

1 Why look past human H3N2

Our reference sequence, `zero_polymer`, is the 566-residue HA0 precursor of A/New York/392/2004 (H3N2), accession `YP_308839.1` (`softmax/code/tests/references.py`; the length and accession are from `FASTA/H3N2/sequences_H3N2_566_homo_segment4.csv` in this repository). So human H3N2 was the obvious natural comparison, and `FASTA/H3N2/site_entropy_566.csv` holds it.

The trouble is that human H3N2 is one virus lineage in one host, drifting antigenically since 1968. A site can be variable there because the fold tolerates variation, or because antibody pressure actively rewards change at that site; and a site can be conserved there because the fold needs it, or just because 50 years is not long enough for anything to have happened. The entropy of one lineage mixes structural constraint with that lineage's history.

Across many HA subtypes, which have been separated for far longer and sit in different hosts, a site that is still conserved is conserved because the protein needs it. That is the quantity the sampler should be reproducing. This is the standard argument for using deep, diverse alignments in sequence-statistics work: the covariation and conservation signal that corresponds to physical constraint emerges from alignments spanning large evolutionary distances [17, 3].

2 The two subtype groups

Influenza A viruses are named by their two surface proteins, HA and neuraminidase (NA): H3N2 means HA subtype 3, NA subtype 2. Eighteen HA subtypes are known, H1 to H18, the last two from bats [28]. A subtype is an antigenic and sequence-divergence category: different subtypes do not cross-react serologically.

The 18 subtypes fall into exactly two phylogenetic clades, called *group 1* and *group 2*. The split is old and shows up in every analysis of HA sequences, going back to the first comparison of all the then known subtypes [1, 18]:

group 1	H1, H2, H5, H6, H8, H9, H11, H12, H13, H16 (and H17, H18, from bats)
group 2	H3, H4, H7, H10, H14, H15

Our H3 reference is in group 2, and H4 and H14 are its closest relatives [1, 18].

What actually differentiates them. All HAs have the same architecture: a membrane-distal globular head built from HA1, which carries the receptor-binding site, on a stalk built mostly from HA2, which carries the fusion machinery [31, 24]. The groups are not different folds; they differ in detail, and the differences that matter are:

- **Sequence divergence.** Between groups, HAs are much further apart than within a group. In this repository's own two-subtype test, the median identity to the H3 `zero_polymer` reference was 0.966 for human H3 sequences and 0.435 for H1 sequences (measured by `ha_site_entropy.py`, which prints it per subtype). Group 1 against group 2 is the deepest division inside influenza A HA [1, 18].
- **Stalk conformation and antibody epitopes.** The clearest functional evidence that the group boundary is real comes from broadly neutralising antibodies that bind the conserved stalk: CR6261 neutralises group 1 HAs but not group 2 [5], CR8020 is the mirror image, binding group 2 and not group 1 [6], and antibodies that cover both groups are rare enough to have been reported as a result in their own right [4]. The two groups present measurably different surfaces in the region that is otherwise the most conserved part of the molecule.
- **Receptor-binding site details.** Group-specific differences in the loops around the receptor-binding site change which sialic acid linkages are preferred and how the site can adapt, which is central to the question of avian-to-human transmission [22, 24].

Why two files and not one. Within group 2, the alignment to `zero_polymer`'s 566 positions is trustworthy at essentially every position, so a group 2 entropy file is the conservative measurement. Adding group 1 roughly triples the evolutionary depth, which is what we want, but the HA1 head alignment gets looser and more columns become unreliable. Building both and comparing them is cheap, so the plan is to build both: disagreement between them is itself informative, and agreement means the result does not hinge on the choice.

3 Influenza B, and why it is excluded

Influenza A and influenza B are different genera of the same family, *Orthomyxoviridae*: *Alphainfluenzavirus* and *Betainfluenzavirus* [12]. The practical differences:

- **Host range and reservoir.** Influenza A has a huge wild bird reservoir, with many subtypes circulating in many hosts, which is what makes its HA so diverse and is the reason it can produce pandemics by reassortment. Influenza B is essentially a human virus, with seals the only other established host [20], so it has nothing like influenza A's diversity.
- **No subtypes, two lineages.** Influenza B is not divided into HA subtypes. It has two antigenically distinct lineages, B/Victoria/2/87-like and B/Yamagata/16/88-like, which diverged in the 1980s [21]. Compared with A's 18 subtypes that is a very shallow set.
- **No reassortment with A.** The two genera do not exchange segments, so their HAs have evolved independently throughout.
- **Same fold, far apart in sequence.** Influenza B HA has the same overall structure as influenza A HA, including the HA1/HA2 cleavage and the fusion peptide [29]. But it is the most distant member of the family from our reference, and this is the problem: a position-by-position mapping onto `zero_polymer`'s 566 sites stops being meaningful when

the alignment itself is uncertain, and the per-site entropy we want is exactly a per-position quantity. The round figure usually quoted for A against B HA identity is about 25%, which is in the range where alignment columns, not biology, would drive the result. *Treat that 25% as approximate: measure it on your own alignment before quoting it in the thesis.*

So influenza B is excluded not because it is irrelevant but because at that distance the mapping step, not the conservation signal, would decide the answer.

4 Where the sequences come from

Three sources are worth knowing. The code only needs FASTA files, so any of them works.

NCBI (the Influenza Virus Resource, now NCBI Virus). The one you already know. It was published as a dedicated influenza resource, with its own sequence set, annotation pipeline and alignment tools [2], and has since been folded into the broader NCBI Virus / Virus Variation platform [10]. It is public, needs no account, and has a programmatic interface, the E-utilities [8], which is what `ha_site_entropy.py fetch` uses: one query per subtype against the protein database, selecting by organism (the taxonomy has a node per HxNy subtype), by protein name, and by sequence length.

BV-BRC (formerly FluDB / IRD). The Influenza Research Database (IRD), whose website was `fludb.org`, was a US-funded influenza-specific resource: sequences plus curated metadata (host, location, date, passage history) and analysis tools, built for surveillance work [25]. In 2019–2022 IRD was merged with PATRIC (bacteria) and ViPR (other viruses) into a single centre, the Bacterial and Viral Bioinformatics Resource Center, BV-BRC, at `bv-brc.org` [19]. So "FluDB"/"IRD"/"BV-BRC" are three names for the same lineage of resource. For us its advantage over NCBI is the curated metadata: if you want "avian H3 only" or "exclude egg-passaged isolates", BV-BRC makes that a filter rather than a text search.

GISAID. The Global Initiative on Sharing All Influenza Data, whose influenza database is EpiFlu [23, 7]. It holds sequences that are not in NCBI, because many national laboratories deposit there under its terms. Those terms are the catch: access requires registration and agreement to a data access agreement that forbids redistribution and requires acknowledging the submitting laboratories. Practically, that means you may use GISAID data in your analysis but you cannot commit the FASTA files to this repository, and the acknowledgement table has to go in the thesis. For our purpose, which is deep subtype coverage rather than the latest circulating strains, NCBI alone is very likely enough, so I would only go to GISAID if some subtype turns out to be thinly covered.

5 Redundancy: clustering and weighting

The problem. Sequence databases are not a sample of the protein's sequence space, they are a sample of what people sequenced. Human H3N2 and H1N1 have been sequenced tens of thousands of times because they infect us; H14 has barely been sequenced at all. Shannon entropy computed on raw counts would therefore mostly describe surveillance effort. Worse, near-identical sequences make every site look conserved: if 10 000 of your 11 000 sequences are the same recent human H3N2 clade, the entropy you measure is that clade's internal diversity.

There are two standard cures.

Clustering: CD-HIT and MMseqs2. Both are programs that take a set of sequences and group them so that no two groups are more similar than a chosen identity threshold, keeping one representative per group. CD-HIT does this with a greedy incremental algorithm: sort by length, take the longest sequence as the first representative, and compare each next sequence against existing representatives, making it a new representative only if it is below the threshold against all of them; short-word filters make the comparisons fast enough to be practical [16, 9]. MMseqs2 is a newer and much faster suite doing the same job (and sensitive search besides), with a linear-time clustering mode for very large sets [26, 27]. Clustering at 90–95% identity is the usual recipe for "remove near-duplicates". The cost is that it throws information away: a cluster of 10 000 sequences and a cluster of 2 both end up as one sequence.

DCA-style weights. The alternative keeps every sequence but downweights the crowded ones. It comes from direct coupling analysis, the Potts-model approach to inferring contacts and fitness from sequence alignments [17, 3]. Each sequence a gets the weight

$$w_a \;=\; \frac{1}{m_a}, \qquad m_a \;=\; \#\{\, b : \text{identity}(a,b) \geq \theta \,\}, \tag{1}$$

where b runs over all sequences including a itself, so $m_a \geq 1$ and $w_a \leq 1$. A sequence with no close neighbours counts fully; each member of a cluster of 100 near-identical sequences counts $1/100$, so the cluster as a whole counts about 1. The effective number of sequences is the total weight,

$$N_{\text{eff}} \;=\; \sum_a w_a, \tag{2}$$

and it is the honest sample size: it says roughly how many independent sequences the alignment is worth. The usual threshold is $\theta = 0.8$, i.e. 80% identity; Morcos *et al.* use 80%, define the weight as the inverse count of sequences above that identity, and define N_{eff} as the sum of the weights, reporting only a weak dependence on the precise threshold [17]. This is what `ha_site_entropy.py` implements, with θ exposed as `-weight-identity`; frequencies and entropies are then computed with w_a in place of a count of 1.

Weighting is the better choice here, and it is what the code does: it is a smooth correction rather than a hard cut, it keeps every sequence, and it hands you N_{eff} as a number you can report, which clustering does not. It is worth knowing that the value of phylogenetic reweighting has been questioned for contact prediction specifically [11]; for our purpose, where the point is precisely not to let surveillance effort set the entropy, the correction is doing something we want regardless.

6 Alignment: what MAFFT is for

Why alignment is needed at all. The H3N2 file needed none: all 19 380 sequences were already 566 residues long, so position 7 meant the same thing in all of them. That stops being true across subtypes. HA sequences differ in length, with insertions and deletions, so the 7th letter of an H5 sequence is not the 7th site of our reference. A multiple sequence alignment inserts gaps so that each column holds positions that are homologous, i.e. descended from the same ancestral position. Only then is a per-column entropy meaningful.

MAFFT. MAFFT is one of the standard multiple sequence alignment programs, originally built around a fast Fourier transform trick for finding homologous segments quickly [13], and in current use as version 7 with a range of accuracy/speed settings [15]. It is free, fast enough for thousands of HA sequences on a laptop (8 seconds for 3 165 sequences in our test), and scriptable. Any comparable program (MUSCLE, Clustal Omega, HMMER's `hmmalign` against the Pfam PF00509 model) would do the same job.

What we align, and to what. All the kept sequences *together with* the `zero_polymer` reference, in one run (`-align msa`). Then, and this is the important step, we keep only the alignment columns where the reference has a residue rather than a gap. That projects the alignment onto the reference's own numbering: insertions relative to `zero_polymer` are discarded, deletions appear as gap characters, and the output has exactly 566 rows, one per site of our reference, which is what the comparison needs. The code checks that the reference comes out of the alignment unchanged, and fails loudly otherwise.

The alternative mode (`-align add`, MAFFT's `-add -keeplength` [14]) aligns each sequence against the reference alone. It is faster and fine within one subtype, which is why it reproduces the existing H3N2 file, but a full multiple alignment is better across subtypes, because the other sequences help place the indels.

Gappy columns. If a column is mostly gaps, the handful of residues in it carry no reliable entropy, and treating "gap" as a 21st amino acid would be worse: it would report alignment uncertainty as biological variability. So the code computes the weighted gap fraction per site and writes `NaN` for the entropy wherever it exceeds `-max-gap` (30% by default), and `structure_zero_polymer.py` excludes those sites from the rank correlation, the overlap counts and the box plots.

7 The one real judgement call: the weighting threshold

This is the long version of the caveat in `ha_natural_runbook.md`.

Take equation (1) with $\theta = 0.8$ and apply it to a cross-subtype set. Within a subtype, sequences are typically far above 80% identical to each other: human H3 sequences sit at 0.966 median identity to our reference, so they are all mutual neighbours. Between subtypes they are far below it: H1 against H3 is about 0.44. The neighbour sets therefore coincide almost exactly with the subtypes, so m_a is about the size of a's subtype, each subtype's weights sum to about 1, and

$$N_{\text{eff}} \approx \text{(number of subtypes)}. \tag{3}$$

Our two-subtype test came out at $N_{\text{eff}} = 2.0$, which looks alarming until you see that it is exactly what the formula is supposed to do.

The consequence is what matters. With these weights the entropy at a site is almost entirely *between*-subtype variation: it asks "do the different subtypes agree at this position?" and is nearly blind to variation inside a subtype, including all of human H3N2's antigenic drift. That is a defensible and arguably the right definition of "what the fold requires", since the within-subtype signal is the part most contaminated by host-specific immune pressure and by sequencing effort. But it is a choice, not a fact, so it should be shown not to matter:

- `-weight-identity 0.8` (the default): one effective sequence per subtype; pure between-subtype conservation.
- `-weight-identity 0.95`: within-subtype drift is kept, since sequences only count as neighbours when they are nearly identical, and each subtype contributes in proportion to how much of it was sequenced. A middle course, and closer to what the original H3N2-only file measured.
- `-weight-identity 0` (no weighting): raw counts, so the entropy is dominated by whichever subtypes have been sequenced most, in practice H1, H3 and H5. This is the sampling-biased extreme and is useful mainly as a control.

Build the file at 0.8 and at 0.95, run the K-site comparison on both, and report that the conclusion is the same. If it is not the same, that is a finding about which kind of conservation the sampler is reproducing, and more interesting than either number alone.

Correction, from the first real run. The paragraph above says N_{eff} comes out near the number of subtypes at $\theta = 0.8$, with each subtype contributing about 1. On the real 16-subtype data that is only true for the subtypes that have no close sister. Sister subtypes are themselves more than 80% identical, so they merge into one cluster, and a small subtype next to a large one is swallowed: H15 (11 unique sequences, sister to H7 with 1 712) came out with $N_{\text{eff}} = 0.0$, and H13 and H16 with 0.7 and 0.4. At $\theta = 0.8$ the entropy is therefore one vote per *cluster of sister subtypes*, not per subtype, and the rarest subtypes drop out of the average altogether. That is a worse definition than intended, so prefer the $\theta = 0.95$ files and keep 0.8 only as a comparison.

8 What the answer should look like

A prediction worth writing down before looking, because it makes the result falsifiable. If the sampler is finding real structural constraint, then in the natural alignment:

- HA2 should be conserved: the fusion peptide at its N-terminus, the long central helix, the stalk. This region does the membrane fusion and is sterically constrained, and it is where the broadly neutralising antibodies bind precisely because it cannot change much [24, 5, 6].
- The HA1 head should be variable, especially the five antigenic sites around the receptor-binding site that were mapped from the original structure and from drift variants [30, 31].

In the repository's two-subtype test this split is already visible: median entropy 1.00 bits over HA1 against 0.084 bits over HA2 (measured by `ha_site_entropy.py`; the HA1/HA2 boundary is found from the fusion peptide motif G[IL]FGAIAGF).

So if the T12 K-sites concentrate in HA2 and the stalk, that is evidence the sampler is doing something physical, before any correlation coefficient is computed. And if they scatter uniformly, or sit in the antigenic sites, no Spearman value will rescue the result.

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